

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. BPT-504: Electronic Devices and Circuits

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer in brief any **five** of the following:

[5 × 4 = 20]

- What is the significance of the threshold voltage (V_T) in an enhancement mode MOSFET? How is its magnitude reduced during fabrication?
- Show that negative feedback improves the stability of the voltage gain of an amplifier against parameter variations.
- Why is a Unijunction Transistor (UJT) not used as an amplifier? Explain the condition under which it is used as a relaxation oscillator.
- What is meant by Common Mode Rejection Ratio (CMRR) in the case of an OP-AMP? Why should its value be ideally infinite?
- Write the Boolean expression represented by the Karnaugh map (K-map) shown below and simplify it:

$A \backslash BC$	00	01	11	10
0	1	1	0	0
1	0	1	1	0

- How is a Cathode Ray Oscilloscope (CRO) superior to ordinary measuring instruments? How does the Sync control help in obtaining a stationary pattern?

Question 2.

- With necessary theory and diagrams, explain the construction, working, and I-V characteristics of a Silicon Controlled Rectifier (SCR). Why can one not use germanium instead of silicon to construct a controlled rectifier? [7]
- Explain the condition under which a UJT exhibits negative resistance. [3.5]
- The firing voltage of a UJT is 15 V. It is connected to a series RC circuit to act as a relaxation oscillator with $R = 50 \text{ k}\Omega$ and $C = 2000 \text{ pF}$. If the source voltage is 40 V, determine the time period of the saw-tooth voltage across the capacitor C . [3.5]

Question 3.

- Refer to the basic common-base BJT amplifier configuration. The h-parameter values for the transistor are:

$$h_{ib} = 40 \Omega, \quad h_{rb} = 3 \times 10^{-4}, \quad h_{fb} = -0.99, \quad h_{ob} = 0.5 \mu\text{mhos}$$

Determine: (i) the closed-loop current gain taking source resistance $R_s = 600\Omega$ and load resistance $R_L = 10\text{ k}\Omega$ into account, and (ii) the input resistance. [6]

- (b) Explain in brief the high-frequency hybrid- π model of a CE transistor and discuss its advantages over the h-parameter model at high frequencies. [8]

OR

- (a) What do you mean by power amplifiers? Give the circuit diagram of class A, class B, and class C power amplifiers. Show that the maximum efficiency of a class B push-pull amplifier is 78.5%. [8]

- (b) What do you mean by non-linear distortion in a power amplifier circuit? Discuss a method to minimize such distortions. [6]

Question 4.

- (a) What do you mean by the term Amplitude Modulation (AM)? Derive the expression for an amplitude-modulated wave and discuss its frequency spectrum. [8]
- (b) Give the truth table of a full adder. Obtain the Boolean expressions for sum (S) and carry (C_o) from this table in SOP form. Simplify these expressions using a K-map and implement the circuit using basic logic gates. [6]

Question 5.

- (a) What is meant by feedback in amplifiers? Differentiate between positive and negative feedback. Draw the circuit diagram of a current-series feedback amplifier and show that negative feedback increases the input resistance and output resistance of the amplifier. [8]
- (b) How does the feedback network of a Hartley oscillator differ from the feedback network of a Colpitts oscillator? What will be the oscillator frequency in the two cases? [6]

OR

- (a) Explain the difference among astable, monostable, and bistable multivibrators in terms of their stable states and applications. [7]
- (b) Define the parameters relative amplification factor (μ), drain resistance (r_d), and transconductance (g_m) in the case of a JFET amplifier, and establish the relationship among them. How is a JFET different from a bipolar junction transistor? [7]